

Aditi Kabra

 [aditink](#) |  [aditink.github.io](#) |  akabra@cs.cmu.edu |  +1 (607) 697 3644

My thesis is about automatically synthesizing verified control envelopes (sets of control solutions) for cyber-physical systems like trains, planes and cars. I'm excited about soundly scaling formal methods techniques using AI.

EDUCATION

Ph.D. Candidate (Computer Science), Carnegie Mellon University, USA 2020 – present

Advisors: Prof. André Platzer and Prof. Stefan Mitsch.

Bachelor of Science, Cornell University, USA 2016 – 2019

magna cum laude in Computer Science, *cum laude* in Mathematics, *cum laude* in Physics.

PUBLICATIONS

Conference Publications

- [CESAR: Control Envelope Synthesis via Angelic Refinements](#). **Aditi Kabra**, Jonathan Laurent, Stefan Mitsch, André Platzer. **TACAS** 2024 .
- [Verified Train Controllers for the Federal Railroad Administration Model: Balancing Competing Brake and Track Forces](#). **Aditi Kabra**, Stefan Mitsch, André Platzer. **EMSOFT** 2022. **Best Paper finalist**.
- [Geometry Types for Graphics Programming](#). Dietrich Geisler, Irene Yoon, **Aditi Kabra**, Horace He, Yinnon Sanders, Adrian Sampson. **OOPSLA** 2020.

Short Papers

- [Can Large Language Models Autoformalize Kinematics?](#) **Aditi Kabra**, Jonathan Laurent, Sagar Bharadwaj, Ruben Martins, Stefan Mitsch, André Platzer. **FMCAD** 2025.

Workshop Papers

- [Online Verification of Commutativity](#). **Aditi Kabra**, Dietrich Geisler, Adrian Sampson. **TAPAS** 2020.

Preprints

- [Hybrid Game Control Envelope Synthesis](#). **Aditi Kabra**, Jonathan Laurent, Stefan Mitsch, André Platzer. arXiv preprint, 2025. <https://doi.org/10.48550/arXiv.2508.05997>.

AWARDS

- Swartz Innovation Commercialization Fellow, 2023.
- Best Paper finalist, EMSOFT 2022.

SERVICE

- External/Sub-reviewer: CADE 2025, NFM 2025, TACAS 2025, FM 2024, HSCC 2024, LICS 2023, NFM 2023, FM 2022, ICCPS 2022, iFM 2022, ICCPS 2021.

- Artifact evaluation program committee: FM 2024, HSCC 2024, ASPLOS 2023, ASPLOS 2021.
- Student Ombudsperson for the CMU Computer Science Department, 2023 – present.
- CMU PLunch Organizer (principles of programming weekly lunch seminar), 2022 – 2023.

TEACHING EXPERIENCE

Teaching Assistant, Carnegie Mellon University:

- Software Foundations of Security & Privacy (Fall 2022)
- Logical Foundations of Cyber Physical Systems (Fall 2021)

Teaching Assistant, Cornell University:

- Data Mining for History and Literature (Fall 2019)
- Introduction to Analysis of Algorithms (Spring 2018, Spring 2019)
- (Consultant) Data Structures and Object Oriented Programming (Spring 2017, Fall 2017)

SYNERGISTIC ACTIVITIES

- *NSF I-Corps (Entrepreneurial Lead)*. Led an effort to develop a commercialization strategy for control envelopes through the NSF I-Corps program. We received \$50,000 in funding and interviewed a hundred industry stakeholders, learning about the market dynamics. The identified adoption barriers and opportunities inform my research priorities.
- *Patent Disclosure*. André Platzer, Stefan Mitsch, Aditi Kabra. Systems and methods for verifying train controllers. U.S. Patent Application No. US 2024/0416981 A1, filed 2 Nov. 2022; published 19 Dec. 2024.
- *Industry Experience*. Software Engineer (IC 2) at Microsoft (2020), Software Engineering Intern at Microsoft (2018, 2019), Engineering Practicum Intern at Google (2017).
- *CodeDay Labs*. In CodeDay Labs 2020 through 2024, I mentored college students to build software engineering projects over the summer in preparation for careers in technology.

REFERENCES

Prof. André Platzer

Alexander von Humboldt Professor
KIT Department of Informatics
Karlsruhe Institute of Technology
Karlsruhe, Germany
platzer@kit.edu

Prof. Eunsuk Kang

Associate Professor
Software and Societal Systems Department
School of Computer Science
Carnegie Mellon University
Pittsburgh, USA
eskang@cmu.edu

Prof. Stefan Mitsch

Associate Professor
School of Computing
DePaul University
Chicago, USA
smitsch@depaul.edu

Prof. Armando Solar-Lezama

Distinguished Professor of Computing
MIT Schwarzman College of Computing
Associate Director and COO of CSAIL
Massachusetts Institute of Technology
Cambridge, USA
asolar@csail.mit.edu